SQL Interview Quiz - Retail DB Challenges

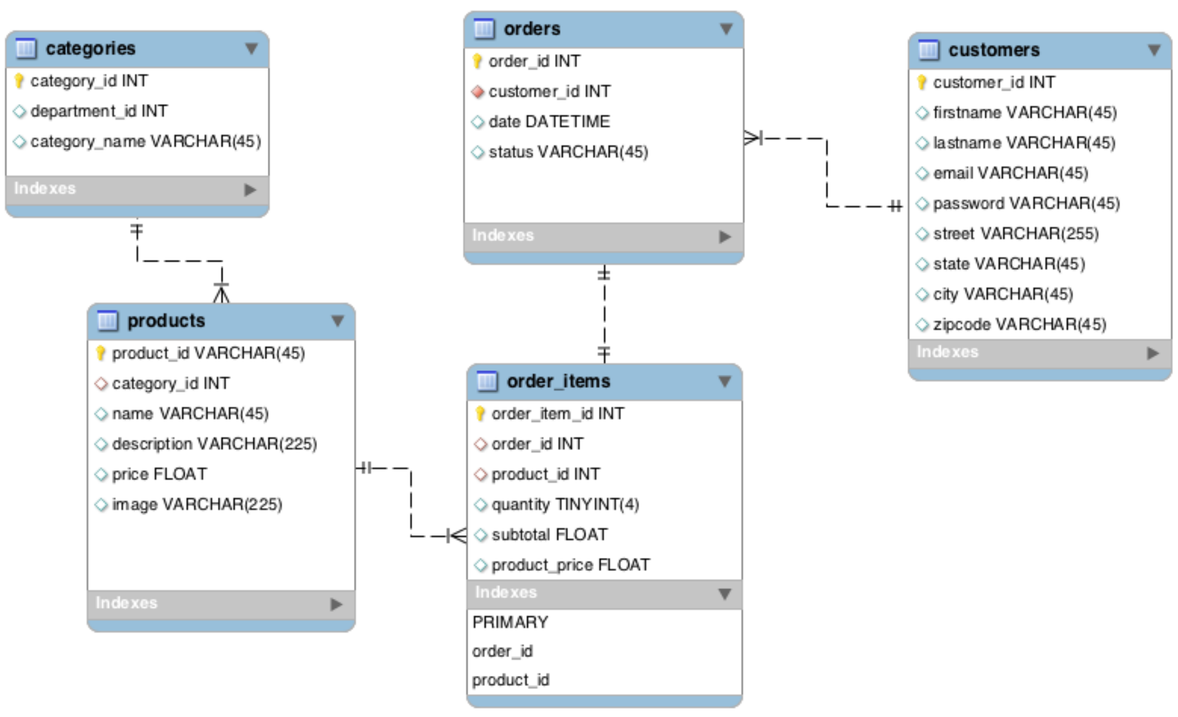
Task description

In this technical interview test, the candidate will be asked to work with a retail database and develop SQL queries to complete 2 questions. The entity relationships among the tables are shown in the graph below. There're 5 tables in the retail database:

- Orders
- Order_items
- Products
- Categories
- Customers

Sample data from the 5 tables are also provided below in the appendix.

Entity Relationship Map



SQL Questions

** Question 1 - Find top 5 states that generated most number of orders?

** Question 2 - Find top 5 product categories (by sales amount) in each city (cities that a have more than 100 consumers)

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# The marketing team wants to launch a campaign and wants to
# find out the top 5 most popular product categories (by sales amount) for each
# city that has more than 100 consumers
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Appendix - Sample Data

department_id	department_name
2	Fitness
3	Footwear
4	Apparel
5	Golf
6	Outdoors
7	Fan Shop

category_id	category_department_id	category_name
1	2	Football
2	2	Soccer
3	2	Baseball & Softball
4	2	Basketball
5	2	Lacrosse
6	2	Tennis & Racquet
7	2	Hockey
8	2	More Sports
9	3	Cardio Equipment
10	3	Strength Training

order_id	order_date	order_customer_id	order_status
1	41480	11599	CLOSED
2	41480	256	PENDING_PAYMENT
3	41480	12111	COMPLETE
4	41480	8827	CLOSED
5	41480	11318	COMPLETE
6	41480	7130	COMPLETE
7	41480	4530	COMPLETE
8	41480	2911	PROCESSING
9	41480	5657	PENDING_PAYMENT
10	41480	5648	PENDING_PAYMENT

order_item_id	order_item_order_id	order_item_product_id	order_item_quantity	order_item_subtotal	order_item_product_price
1	1	957	1	299.98	299.98
2	2	1073	1	199.99	199.99
3	2	502	5	250	50
4	2	403	1	129.99	129.99
5	4	897	2	49.98	24.99
6	4	365	5	299.95	59.99
7	4	502	3	150	50
8	4	1014	4	199.92	49.98
9	5	957	1	299.98	299.98
10	5	365	5	299.95	59.99

customer_id	customer_fname	customer_lname	customer_email	customer_password	customer_street	customer_city	customer_state	customer_zipcode
1	Richard	Hernandez	XXXXXXXXX	XXXXXXXXX	6303 Heather Plaza	Brownsville	TX	78521
2	Mary	Barrett	XXXXXXXXX	XXXXXXXXX	9526 Noble Embers Ridge	Littleton	CO	80126
3	Ann	Smith	XXXXXXXXX	XXXXXXXXX	3422 Blue Pioneer Bend	Caguas	PR	725
4	Mary	Jones	XXXXXXXXX	XXXXXXXXX	8324 Little Common	San Marcos	CA	92069
5	Robert	Hudson	XXXXXXXXX	XXXXXXXXX	10 Crystal River Mall	Caguas	PR	725
6	Mary	Smith	XXXXXXXXX	XXXXXXXXX	3151 Sleepy Quail Promenade	Passaic	NJ	7055
7	Melissa	Wilcox	XXXXXXXXX	XXXXXXXXX	9453 High Concession	Caguas	PR	725
8	Megan	Smith	XXXXXXXXX	XXXXXXXXX	3047 Foggy Forest Plaza	Lawrence	MA	1841
9	Mary	Perez	XXXXXXXXX	XXXXXXXXX	3616 Quaking Street	Caguas	PR	725
10	Melissa	Smith	XXXXXXXXX	XXXXXXXXX	8598 Harvest Beacon Plaza	Stafford	VA	22554

product_id	product_category_id	product_name	product_description	product_price	product_image
1	2	Quest Q64 10 FT. x 10 FT. Slant Leg Instant U		59.98	http://images.acmesports.sports/Quest+Q64+10+FT.+x+10+FT.+Slant+Leg+Instant+Up+Canopy
2	2	Under Armour Men's Highlight MC Football Clea		129.99	http://images.acmesports.sports/Under+Armour+Men%27s+Highlight+MC+Football+Cleat
3	2	Under Armour Men's Renegade D Mid Football Cl		89.99	http://images.acmesports.sports/Under+Armour+Men%27s+Renegade+D+Mid+Football+Cleat
4	2	Under Armour Men's Renegade D Mid Football Cl		89.99	http://images.acmesports.sports/Under+Armour+Men%27s+Renegade+D+Mid+Football+Cleat
5	2	Riddell Youth Revolution Speed Custom Footbal		199.99	http://images.acmesports.sports/Riddell+Youth+Revolution+Speed+Custom+Football+Helmet
6	2	Jordan Men's VI Retro TD Football Cleat		134.99	http://images.acmesports.sports/Jordan+Men%27s+VI+Retro+TD+Football+Cleat
7	2	Schutt Youth Recruit Hybrid Custom Football H		99.99	http://images.acmesports.sports/Schutt+Youth+Recruit+Hybrid+Custom+Football+Helmet+2014
8	2	Nike Men's Vapor Carbon Elite TD Football Cle		129.99	http://images.acmesports.sports/Nike+Men%27s+Vapor+Carbon+Elite+TD+Football+Cleat
9	2	Nike Adult Vapor Jet 3.0 Receiver Gloves		50	http://images.acmesports.sports/Nike+Adult+Vapor+Jet+3.0+Receiver+Gloves
10	2	Under Armour Men's Highlight MC Football Clea		129.99	http://images.acmesports.sports/Under+Armour+Men%27s+Highlight+MC+Football+Cleat